

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Aim

To implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis from a given set of training examples.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- Pandas

Procedure

1. Create a training dataset containing positive and negative examples.
2. Initialize the hypothesis with the most specific values.
3. Consider only positive training examples.
4. Compare each positive example with the current hypothesis.
5. Generalize the hypothesis whenever required.
6. Display the final most specific hypothesis.

Code:

```
import pandas as pd

data = {
    'Sky': ['Sunny', 'Sunny', 'Rainy', 'Sunny'],
    'AirTemp': ['Warm', 'Warm', 'Cold', 'Warm'],
    'Humidity': ['Normal', 'High', 'High', 'High'],
    'Wind': ['Strong', 'Strong', 'Strong', 'Strong'],
    'Water': ['Warm', 'Warm', 'Warm', 'Cool'],
    'Forecast': ['Same', 'Same', 'Change', 'Change'],
    'EnjoySport': ['Yes', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)

hypothesis = ['0'] * 6

for i in range(len(df)):
```

```

if df.iloc[i, -1] == 'Yes':
    for j in range(6):
        if hypothesis[j] == '0':
            hypothesis[j] = df.iloc[i, j]
        elif hypothesis[j] != df.iloc[i, j]:
            hypothesis[j] = '?'
print("Training Data:")
print(df)
print("\nFinal Most Specific Hypothesis:")
print(hypothesis)

```

Output:

```

print(hypothesis)

```

Training Data:

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes

Final Most Specific Hypothesis:
 ['Sunny', 'Warm', '?', 'Strong', '?', '?']

Result

Thus, the **FIND-S algorithm** was successfully implemented and the most specific hypothesis was obtained.